As shown in the figure, the orbitals lying along the axes dx^2-y^2 and dz^2 orbitals will experience strong repulsion and raise in energy to a greater extent than the orbitals with lobes directed between the axes (d_{xy}, d_{yz} and d_{zx}). Thus the degenerate d orbitals now split into two sets and the process is called crystal field splitting.

Step 3: Up to this point the complex formation would not be favoured. However, when the ligands approach further, there will be an attraction between the negatively charged electron and the positively charged metal ion, that results in a net decrease in energy. This decrease in energy is the driving force for the complex formation.

Crystal field splitting in octahedral complexes:

During crystal field splitting in octahedral field, in order to maintain the average energy of the orbitals (barycentre) constant, the energy of the orbitals $d_{x^2-y^2}$ and d_{z^2} (represented as e_g orbitals) will increase by $3/5\Delta_o$ while that of the other three orbitals d_{xy}, d_{yz} and d_{zx} (represented as t_{2g} orbitals) decrease by $2/5\Delta_o$. Here, Δ_o represents the crystal field splitting energy in the octahedral field.

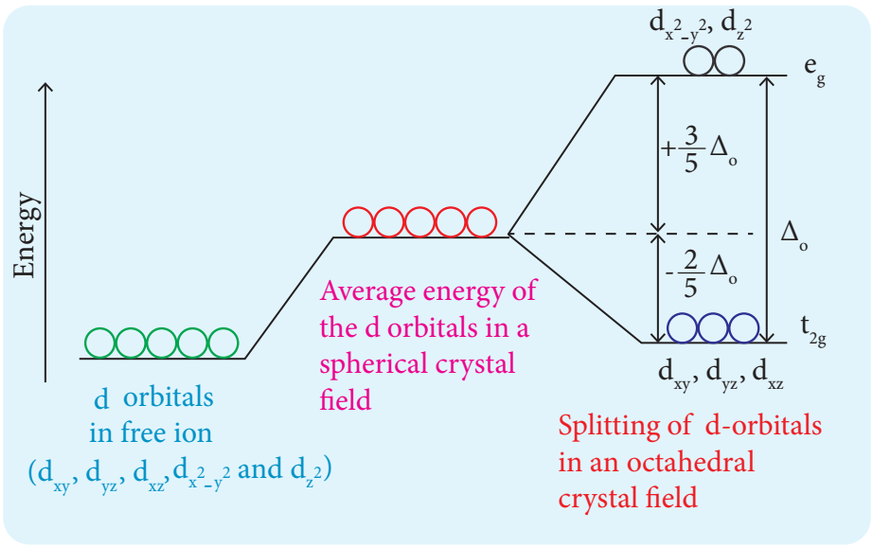


Figure: 5.10 - Crystal field splitting in octahedral field

Crystal field splitting in tetrahedral complexes:

The approach of ligands in tetrahedral field can be visualised as follows. Consider a cube in which the central metal atom is placed at its centre (i.e. origin of the coordinate axis as shown in the figure). The four ligands approach the central metal atom along the direction of the leading diagonals drawn from alternate corners of the cube.

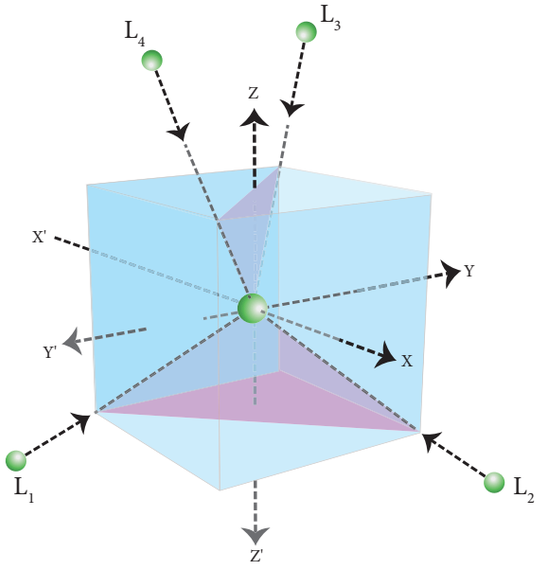


Figure 5.11 tetrahedral ligand field

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